MECHANISTIC INTERPRETABILITY · RESEARCH ARTICLE

Investigating “John Gave a Bag to John”

A Preliminary Study of Repeated-Name Sensitivity and Candidate Circuits in GPT-2 Medium

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Code and project materials
github.com/fahdev789/activation-patching-gpt2

Model focus: GPT-2 Medium (24 transformer blocks; 355M parameters in the supplied notes)
Research status: Preliminary exploratory analysis; causal circuit identification not yet established
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This manuscript integrates the supplied experimental notes with the publicly accessible README and figures in the linked repository. Reported measurements are attributed to their respective source. Repository notebooks and raw activation artifacts were not independently executed or audited for this manuscript.

Abstract

When a proper name occurs twice in one sentence, the two positions share lexical content but may play different grammatical roles. This paper presents a preliminary mechanistic-interpretability investigation of the sentence “John gave a bag to John,” focusing on whether GPT-2 Medium exhibits repeated-name-sensitive attention and context-dependent representations. The supplied exploratory notes report (i) distinct GPT-2 byte-pair token IDs for the two occurrences of “John” (7554 and 1757), (ii) residual-stream cosine similarity reaching 0.8869 at Layer 12 and rising to 0.9541 by Layer 23, and (iii) a strong match-condition attention contrast for Layer 1, Head 4 (L1H4): 0.4175 attention from the recipient position to the subject position in the repeated-name prompt, versus 0.0189 and 0.0107 in two mismatch controls.

The linked repository adds a second, related line of investigation. Its identical-entity-circuits README describes a yes/no prompt asking whether giver and recipient are the same, and publishes figures labelled as token-level causal attribution / patched-logit-difference analyses. The repository figure reports a second-John column peaking at 2.141 at Layer 2 and declining to 1.579 at Layer 23, against a displayed baseline of approximately 1.561. These values are transcribed from the published figure; the README does not provide enough methodological detail to independently reconstruct the intervention or fully interpret the metric.

The combined evidence nominates repeated-name sensitivity as a worthwhile target for controlled experiments, but does not establish that L1H4 tracks entities or that a complete causal circuit has been identified. Lexical-match sensitivity, role-conditioned representation, and entity-level disambiguation are distinct hypotheses. The paper formalizes these distinctions, presents the reported results, and specifies reproducibility and intervention requirements for future work.

Keywords: mechanistic interpretability; GPT-2 Medium; repeated names; entity representation; attention heads; residual stream; activation patching; causal circuits

Contributions and scope

1. Consolidates preliminary attention and representation probes for repeated-name prompts.
2. Incorporates the linked repository’s publicly documented yes/no task and token-patching figure as a distinct evidence stream.
3. Separates observed numerical patterns from hypotheses about entity tracking and causal mechanisms.
4. Provides equations, figure captions, limitations, and a reproducibility checklist for follow-up work.

Evidence boundary. The attention-head values and residual-similarity endpoints come from the supplied research notes. The layerwise patched-logit values come from visual transcription of the repository figure. These are not a single unified experiment and should not be merged as if they shared an identical protocol.

1. Introduction

Mechanistic interpretability seeks explanations of model behavior in terms of internal computational components and the information they transmit [1,2]. A well-studied example is indirect-object identification (IOI), in which GPT-family models use contextual cues to select a recipient name. Prior work has shown that IOI behavior can depend on multiple interacting attention-head classes, and that credible circuit explanations require more than identifying a visually salient head [1].

The present study asks a narrower question: what happens when the subject and recipient positions contain the same written name? In “John gave a bag to John,” the first occurrence is the giver and the second is the recipient. The sentence is intentionally under-specified with respect to whether the two mentions refer to one person or two distinct people with the same name. That ambiguity makes the prompt useful for distinguishing surface-form matching from deeper referential interpretation.

1.1 Research questions

RQ1. Do the two name positions become representationally differentiated across GPT-2 Medium layers?
RQ2. Does any attention head exhibit a repeated-name-dependent attention pattern from the recipient occurrence to the subject occurrence?
RQ3. Do interventions on token or component activations alter a defined same-person/different-person behavioral readout?
RQ4. Does the observed behavior generalize across names, templates, tokenization patterns, and explicit identity context?

1.2 Hypotheses

H1 — Contextual differentiation: representations at the two positions diverge as role and context information are incorporated.
H2 — Lexical repetition sensitivity: one or more heads respond more strongly when the subject and recipient names match.
H3 — Entity disambiguation: the model distinguishes same-name mentions referring to one person from same-name mentions referring to two people. This is the strongest claim and requires direct behavioral and causal evidence.

2. Background and formalization

2.1 Transformer attention

For a standard attention head, the attention distribution from query position i to key position j is computed from query and key vectors. In simplified notation:

$$A_{ij} = \mathrm{softmax}_j((q_i \cdot k_j) / \sqrt{d_k})$$

A high value A_{ij} indicates that position i assigns substantial attention weight to position j within that head. It does not alone establish what information is carried by the corresponding value vector, how the output projection transforms it, or whether downstream components use it. This distinction is central to interpreting the L1H4 result [3,4].

2.2 Residual-stream similarity

Let $r_{l,p}$ denote the residual-stream vector at layer l and token position p. The cosine similarity between the two name positions is:

$$\cos(r_{l,1}, r_{l,2}) = (r_{l,1} \cdot r_{l,2}) / (||r_{l,1}|| \, ||r_{l,2}||)$$

Cosine similarity measures angular alignment, not semantic identity. It can be affected by shared lexical content, role information, layer normalization, and unrelated contextual features. Interpretation therefore requires baselines and a precisely specified hook point.

2.3 Activation patching

Activation patching compares a clean run with a corrupted run and replaces an activation in the corrupted computation with the corresponding activation from the clean computation. A behavioral metric is then measured before and after the intervention [4,5]. For a logit-difference metric:

$$D(x) = \mathrm{logit}_x(y_{target}) - \mathrm{logit}_x(y_{foil})$$

A common normalized recovery score is:

$$\text{Recovery} = (D_{\text{patched}} - D_{\text{corrupted}}) / (D_{\text{clean}} - D_{\text{corrupted}})$$

This equation is included as a recommended reporting convention. The README figure in the linked subdirectory labels its plotted quantity as patched logit difference, but does not fully specify the donor/corrupted pair or normalization. The plotted values are therefore not relabelled as normalized recovery.

3. Materials and methods

3.1 Model and prompt family

The supplied notes identify GPT-2 Medium (355M parameters; 24 layers). Three minimal sentence conditions were used for the attention comparison:

Condition	Prompt	Contrast
S1 (match)	John gave a bag to John.	Same name at subject and recipient.
S2 (mismatch A)	John gave a bag to Michael.	Subject name held fixed; recipient differs.
S3 (mismatch B)	Michael gave a bag to John.	Recipient name held fixed; subject differs.

3.2 Tokenization

The notes report token ID 7554 for the first “John” and 1757 for the second. This is consistent with GPT-2’s whitespace-sensitive byte-pair encoding, where a leading-space form can differ from a sequence-initial form. These IDs should be rechecked against the exact tokenizer version and literal prompt strings. The two positions are not identical input tokens, despite sharing the same visible name.

3.3 Measurements in the supplied notes

(a) Cosine similarity between the residual-stream vectors at the two name positions across the 24-layer network. Only two numerical landmarks are specified in the supplied notes: the minimum at Layer 12 and the Layer 23 value.
(b) Attention pattern at the second name position directed to the first name position, using the reported TransformerLens hook path blocks.{layer}.attn.hook_pattern. L1H4 was highlighted.
(c) The notes mention L23H5 as comparatively stable across the three prompts, but provide no numeric values for it.

3.4 Repository evidence reviewed

The linked subdirectory README describes an additional prompt: “John gave bag to John: Are the recipient and the giver same?” with “yes” or “no” as candidate final tokens. It explicitly cautions that the work is incomplete/unfinished and raises tokenization and positional-attention questions. Its two published figures are labelled as a circuit/causal-attribution visualization and a layer-by-token causal-attribution heatmap. The latter displays a baseline near 1.561 and a second-John column with values from 2.130 at Layer 0 to 1.579 at Layer 23, peaking at 2.141 at Layer 2.

The parent repository README documents separate GPT-2 activation-patching notebooks, including GPT-2 Small dog/cat residual patching and a GPT-2 Medium Layer-4 attention-versus-MLP recovery test. Those are relevant project infrastructure and methodological context, but they are not the same-name experiment and are not treated here as direct evidence for entity disambiguation [6].

3.5 Analysis limitations

The repository files were reviewed through their public README and displayed PNG figures. The notebooks, code, model outputs, activation caches, and environment were not executed in this report. Consequently, repository-figure values are described as figure-transcribed values, not independently reproduced measurements. The plotted heatmap values are rounded to three decimals in the source figure.

4. Results

4.1 Tokenization result

The two visible “John” mentions reportedly map to different token IDs: 7554 at the first position and 1757 at the second. This is an important control issue: any observed repeated-name effect could be influenced by the distinction between sequence-initial and leading-space token embeddings. A robust test should include name variants and position-matched tokenization controls.

4.2 Residual-stream similarity

The supplied notes report that similarity decreases through early layers, reaches its lowest measured value of 0.8869 at Layer 12, and increases to 0.9541 at Layer 23. Intermediate numeric values were not supplied, so Figure 1 intentionally plots only the two documented points rather than inventing a full trajectory.

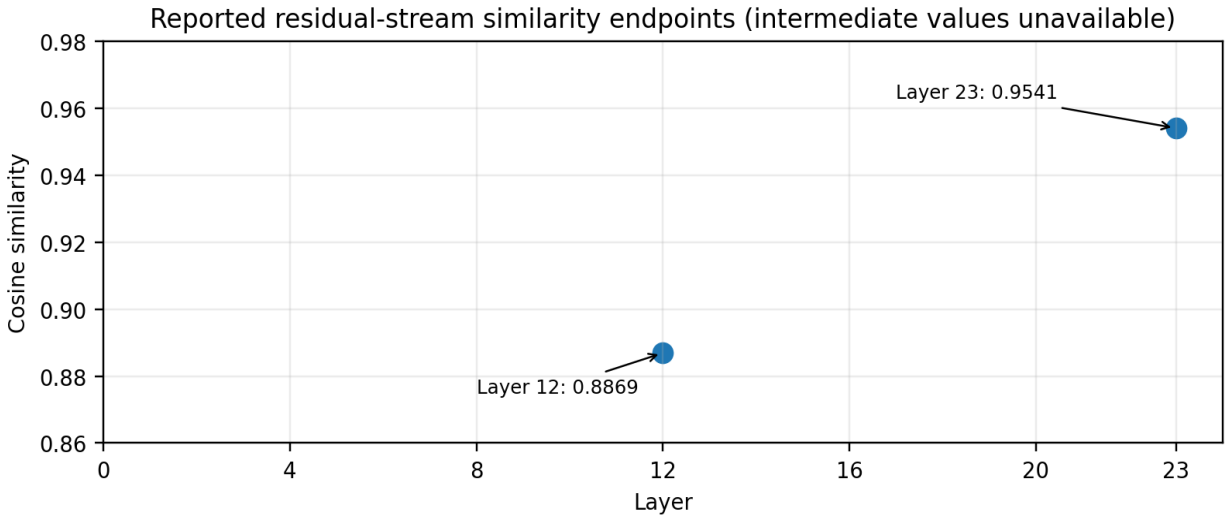


Figure 1. Reported residual-stream cosine-similarity landmarks for the first and second name positions. Only the Layer 12 minimum and Layer 23 endpoint are numerically available in the supplied notes; no intermediate values are interpolated.

A plausible hypothesis is that intermediate layers add role- and context-specific information, separating the two positions, while later layers yield more aligned representations. However, the reported endpoint similarity remains high, and no baseline distribution is available. This result should be treated as a localization clue, not a direct measure of distinct-entity encoding.

4.3 Attention-head comparison: L1H4

The reported L1H4 attention from the recipient position to the subject position is 0.4175 in S1, 0.0189 in S2, and 0.0107 in S3. The repeated-name condition is therefore substantially higher than either mismatch condition in this three-prompt comparison.

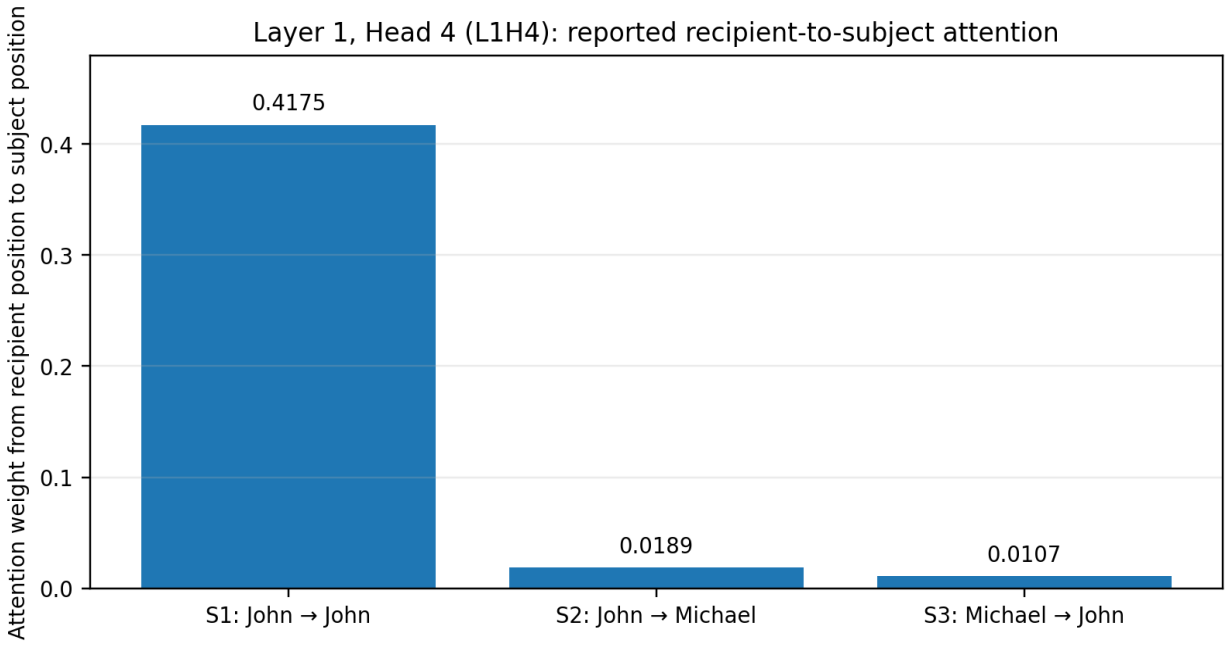


Figure 2. Layer 1, Head 4 attention weight from the recipient-position query to the subject-position key for the three supplied sentence conditions. Values are reproduced from the supplied notes. These are attention weights, not causal effect sizes.

The contrast makes L1H4 a strong candidate for follow-up. However, the result does not distinguish a general repeated-name detector from token-specific matching, positional effects, or another correlated feature. No confidence intervals or across-name replication statistics were supplied.

4.4 Repository token-patching figure

The linked repository's identical-entity-circuits figure reports a per-layer column for the second "John" (John■), with a displayed baseline of approximately 1.561. The values transcribed from the figure are shown in Figure 3. The John■ value is highest at Layer 2 (2.141), then generally declines toward Layer 23 (1.579). The other token columns in the figure remain close to the displayed baseline, with smaller deviations.

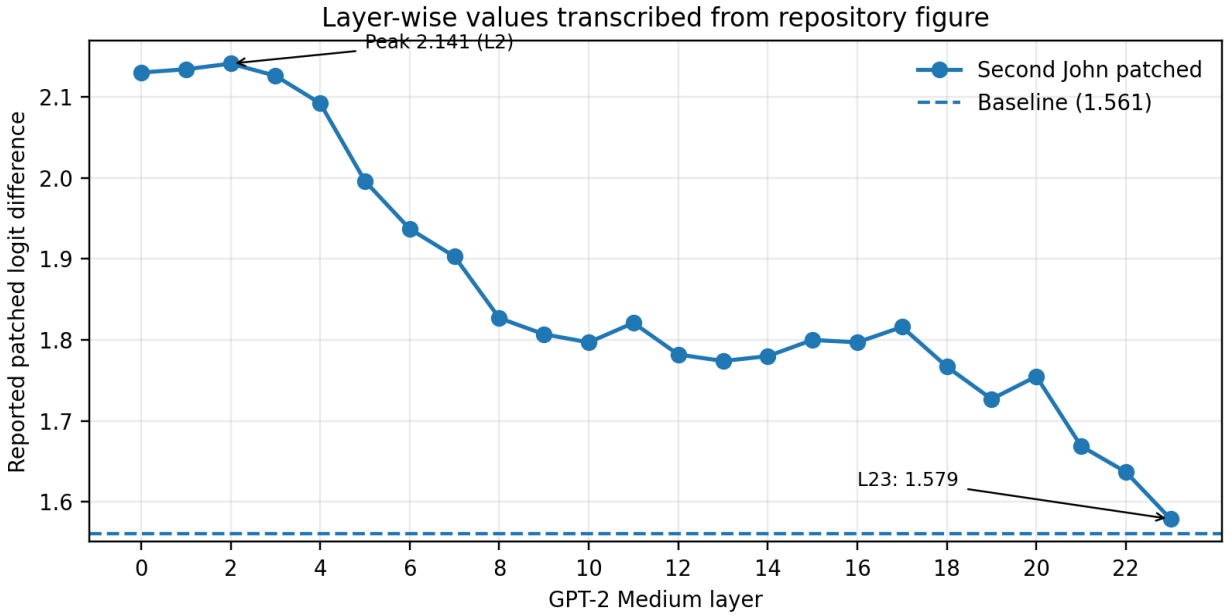


Figure 3. Layer-wise values transcribed from the repository's "Causal Attribution of Key Tokens across Layers" heatmap. The dashed baseline is 1.561, as displayed in the figure. The source README does not fully specify the intervention construction or whether this quantity is normalized recovery; it is therefore labelled only as the reported patched logit difference/attribution value.

This repository result is related to, but methodologically distinct from, the L1H4 attention comparison. It suggests that the second-name token is influential under the repository's reported token-level intervention/attribution setup, especially in early layers. It does not identify a specific attention head, nor does it by itself show that the model correctly distinguishes one person from two people with the same name.

4.5 Summary of evidence streams

Evidence stream	Reported observation	What it supports	What it does not prove
Tokenization	John■=7554; John■=1757	Whitespace-sensitive token distinction.	No conclusion about entity representation.
Residual similarity	0.8869 at L12; 0.9541 at L23	Layerwise representational geometry may change.	No proof of role or referent disentanglement.
L1H4 attention	0.4175 vs 0.0189 / 0.0107	Strong match-sensitive attention pattern in tested prompts.	No proof of causal function or entity tracking.
Repo token figure	John■ peaks at 2.141 (L2); 1.579 (L23)	Token-level intervention/attribution warrants replication.	No independently verified protocol or head-level circuit.

5. Discussion

5.1 Lexical repetition versus entity tracking

The strongest direct interpretation of the L1H4 observation is that the head's recipient-to-subject attention is sensitive to the repeated-name condition in the three prompts tested. This is a lexical/structural observation. Entity tracking is a stronger representational claim: the model would need to distinguish the referents associated with two mentions, potentially recognizing that two people can share the same name. The current prompt set does not establish that distinction.

The yes/no prompt in the repository is a useful move toward behavioral evaluation, but the question "Are the recipient and the giver same?" is itself ambiguous without an explicit discourse context. The sentence "John gave a bag to John" can naturally describe one person giving a bag to themselves, or two different people with the same name. A model response is therefore not an unambiguous ground-truth test of entity disambiguation unless the prompt explicitly defines the referents.

5.2 Attention is routing, not explanation

Attention weights are one part of a head's computation. To test whether L1H4 transmits a repeated-name feature, interventions should target the head output (and, where appropriate, its per-position contribution), not only its attention pattern. Downstream logits and role-sensitive behavior should be measured. A high attention weight may be epiphenomenal, redundant, or part of a computation whose effect is mediated by other heads and MLPs [1,4,5].

5.3 Interpreting the repository's token-patching plot

The repository plot suggests that patching/attributing the second "John" has a larger displayed effect than patching several other listed tokens, particularly in early layers. But a single token-level intervention does not identify a minimal circuit. The interpretation depends on the exact clean/corrupted prompt pair, donor activation, patch location, target and foil tokens, and whether the plotted quantity is a raw logit difference or normalized recovery. These details should be stated in the notebook and reported alongside the figure.

5.4 Relation to IOI circuit research

The IOI literature provides a methodological precedent for treating a circuit as a set of interacting components whose roles are tested by interventions, rather than inferred from attention maps alone [1]. The present task is not identical to IOI: it concerns repeated names in an explicit subject–recipient construction, and its core ambiguity is whether two mentions denote one or two entities. Established IOI head labels should not be transferred to L1H4 or other components without direct evidence.

6. Limitations and threats to validity

Threat	Implication
Prompt count and diversity	Three minimal prompts are insufficient to establish generalization across names, verbs, syntactic forms, and contexts.
Tokenization confound	The two name positions reportedly use different BPE token IDs; leading-space and token-frequency effects need controls.
Ambiguous target behavior	The bare repeated-name sentence does not specify whether the referents are one person or two people.
Measurement specification	The residual hook point, attention tensor indexing, averaging procedure, and precise patching operation need documentation.
No uncertainty estimates	The supplied attention and similarity values have no reported standard errors, confidence intervals, or replication counts.
Causal interpretation gap	Attention patterns and representational similarity are observational; the repository figure's patching protocol is not fully documented in its README.
No independent reproduction	Repository notebooks and raw activations were not executed or audited for this paper.

7. Proposed experiments

7.1 Generalization across names and tokenization

Test a diverse set of first names, including single-token and multi-token names, with matched and mismatched subject–recipient pairs. Balance sequence-initial and leading-space forms, name frequency, and position. Include repeated non-name tokens as controls. Report per-prompt L1H4 attention distributions and confidence intervals.

7.2 Explicit same-person versus two-person contexts

Construct minimally different prompts that explicitly establish the referents. For example: (A) “John■, the same person, gave a bag to himself”; (B) “John■ gave a bag to another person named John■.” Add follow-up questions with unambiguous labels (e.g., “Did Person A give the bag to Person A or Person B?”). This makes the behavioral target identifiable rather than relying on the ambiguous bare sentence.

7.3 Causal intervention on L1H4

Run clean and corrupted prompts that differ only in the name-match feature while preserving syntactic roles and token positions. Cache L1H4 outputs and patch them at the recipient position, then compare with ablation and random-head controls. Evaluate changes in logit difference for explicit behavioral targets. Report both raw logit changes and normalized recovery, including the clean/corrupted denominators.

7.4 Path and circuit reconstruction

Use path patching or automated circuit discovery to test which downstream heads and MLPs receive the candidate signal. Evaluate faithfulness (does the proposed circuit reproduce the behavior?), completeness (does it account for the behavior?), and minimality (can components be removed without loss?)—criteria emphasized in prior circuit work [1,5].

7.5 Conceptual experimental pipeline

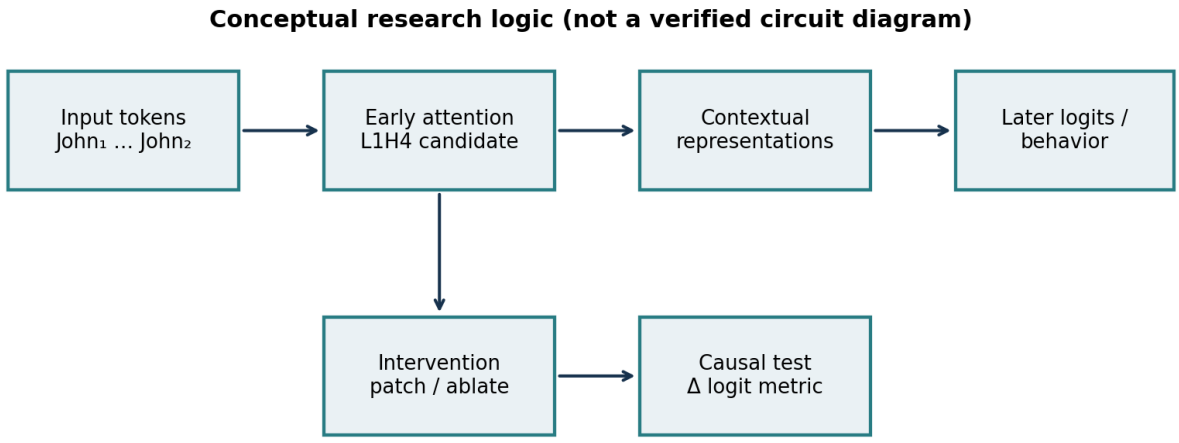


Figure 4. Proposed analysis logic: token-level observations nominate candidate components; controlled interventions test whether those components causally affect a defined behavioral metric. Arrows are conceptual and do not depict an empirically established circuit.

8. Reproducibility and reporting checklist

A reproducible release should include the following artifacts and metadata:

1. Exact prompt strings, punctuation, capitalization, whitespace, and token position annotations.

2. Tokenizer name/version and complete token IDs plus decoded token strings for every prompt.

3. Model identifier, checkpoint revision, library versions, device, random seeds, and deterministic settings.

4. Exact hook names, tensor shapes, residual-stream stage (pre/post block), head index convention, and position indexing.

5. Clean/corrupted prompt construction, donor activation source, patch location, and whether patching targets hook_z, hook_result, resid_pre, resid_mid, resid_post, or another tensor.

6. Behavioral target and foil tokens, logit-difference definition, raw logits, and normalization formula.

7. Per-prompt measurements, summary statistics, uncertainty intervals, and exclusion criteria.

8. Controls: random heads, unrelated positions, token-frequency-matched names, multi-token names, and prompt-template holdouts.

9. Code to regenerate each figure from saved numerical arrays; do not rely on values manually transcribed from screenshots.

10. Raw activation caches or compact derived artifacts sufficient for independent verification, subject to repository size constraints.

9. Conclusion

The preliminary results point to a potentially interesting repeated-name-sensitive pattern in GPT-2 Medium. The supplied notes report a marked L1H4 attention contrast in the repeated-name condition and a non-monotonic change in residual-stream similarity between the two name positions. The linked repository adds a yes/no same-entity prompt and a token-level layerwise attribution figure in which the second “John” has the largest displayed early-layer value.

The evidence currently supports a cautious conclusion: repeated-name prompts are a promising probe for studying the interaction between lexical identity, grammatical role, and contextual representations. It does not yet establish that the model represents two distinct same-name entities, that L1H4 is causally responsible for entity disambiguation, or that a complete circuit has been mapped. The decisive next step is to make the referential task unambiguous and conduct controlled, replicated interventions with clearly defined behavioral metrics.

10. Data and code availability

Project repository: https://github.com/fahdev789/activation-patching-gpt2
Identical-entity-circuits materials: identical-entity-circuits
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The subdirectory README describes the experiment as incomplete/unfinished. The repository’s published figures are included here as reported materials, with the layerwise values in Figure 3 transcribed from the displayed heatmap. Readers should consult the repository for the latest code and artifacts; the present manuscript does not claim to have independently reproduced the experiments.

11. References

[1] K. Wang, A. Variengien, A. Conmy, B. Shlegeris, and J. Steinhardt. “Interpretability in the Wild: A Circuit for Indirect Object Identification in GPT-2 Small.” ICLR, 2023. arXiv:2211.00593.

[2] N. Elhage et al. “A Mathematical Framework for Transformer Circuits.” Transformer Circuits, 2021. Online article.

[3] C. Olsson et al. “In-context Learning and Induction Heads.” 2022. arXiv:2209.11895.

[4] F. Zhang and N. Nanda. “Towards Best Practices of Activation Patching in Language Models: Metrics and Methods.” ICLR, 2024. arXiv:2309.16042.

[5] A. Conmy, A. N. Mavor-Parker, A. Lynch, S. Heimersheim, and A. Garriga-Alonso. “Towards Automated Circuit Discovery for Mechanistic Interpretability.” NeurIPS, 2023. arXiv:2304.14997.

[6] F. (@fahdev789). “Activation Patching & Logit Lens on GPT-2.” GitHub repository README, accessed 8 November 2026. Repository README.

[7] F. (@fahdev789). “Finding Duplicate Detector Circuits in GPT-2.” GitHub subdirectory README and accompanying figures, accessed 8 November 2026. Identical-entity-circuits directory.

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